Serfis

LEAD-GEN AGENTS / DASHBOARD

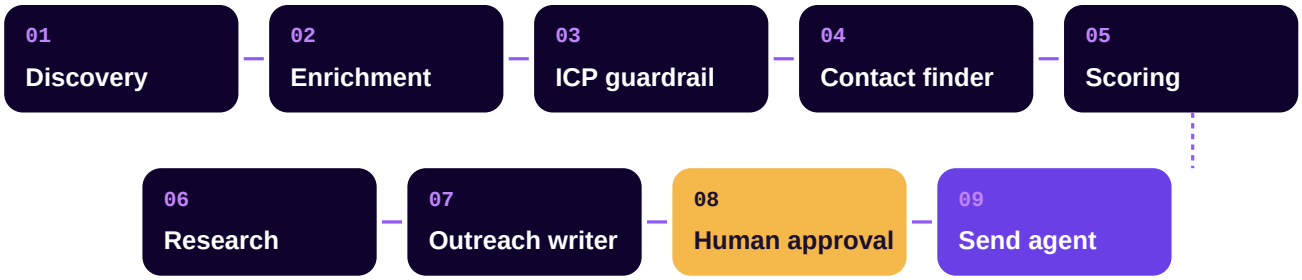
How our agents and dashboard work

A plain-language guide to the nine agents, the review dashboard, and the guardrails that keep every email human-approved.

SERFIS AI / UTD CAPSTONE / OCTOBER 2026

1. What the system does

Serfis sells short-form video ads and social content to early-stage tech startups. This system finds those startups, checks that they fit, finds the right person to contact, researches them and drafts a personal first email. **A person approves every email before anything is sent.**



Steps 1 to 7 run automatically with one command. Step 8 is you, in the dashboard. Step 9 only sends what you approved.

> **Where things stand today:** 7 leads are in the database (6 pending review, 1 fictional sample). Gmail sending is connected. The Claude API key is **not** set, so scoring, research and email drafts use the built-in rule-based text. The mailing address needed in every email is **not** set yet, so real sends are blocked until it is.

How a company becomes a lead

- **Discovery** picks candidates from the YC prospect tracker, then from recent YC batches.
- **Enrichment** adds size, industry, location and founders.
- The **ICP guardrail** rejects anything that does not fit, before any money is spent.
- The **Contact finder** must find a decision-maker's email. If it cannot, the company is cancelled and never saved.
- Only then are the lead scored, researched and given a draft email, and saved for your review.

The orchestrator

`orchestrator.py` runs the agents in this order for each company and saves after every step. It also keeps a report of what was rejected (with reasons), cancelled for no email, or failed, which `main.py run` prints at the end.

2. The agents

Each agent has one job. Below: what it does, how it decides, and the settings that control it.

01 Discovery agent `agents/discovery.py, yc_pool.py, yc.py`

Job: Finds companies that match the ideal customer.

- Reads `data/target_companies.csv` (your YC F26 Prospect Tracker) first: priority P1, then P2. P3 research-only companies and unreviewed rows are never used.
- Skips companies that are already leads or were already looked up, so it never pays twice.
- When the tracker is used up, it draws from recent YC batches (Fall 2026 back to Summer 2025, from the open yc-oss dataset), screens them with the ICP rules for free, then checks each company's own site for a published founder email.
- Ranks candidates by what earlier lookups taught it (batch, team size and market hit rates); larger and hiring companies go first.
- Demo mode (no Apify token) uses a small seed list instead.

Output: a list of candidate companies with their website, YC profile and tracker details.

02 Enrichment agent `agents/enrichment.py, classify.py`

Job: Fills in the facts the other agents depend on.

- Company size band (1-10, 11-50, 51-200 and so on), industry from a fixed list, location and what they sell.
- From the YC profile: team size, location, batch and the founders with their titles. These facts beat anything guessed from a website.
- Uses rule-based detection; uses Claude instead when an API key is set.

Output: a fully described company ready for the guardrail and contact finder.

03 ICP guardrail `agents/icp.py`

Job: Decides whether a company is allowed to become a lead.

- Tracker companies must be priority P1 or P2 and rated Medium fit or better. Research-only comparables are excluded.
- Other companies must show startup evidence (a YC page, a small team, or funding wording) and a clear consumer or B2B fit.
- Rejects developer tools and infrastructure (for example SDKs), services firms and anything over 200 people.
- Fails closed: if the criteria cannot be verified, the company is rejected. Each rejection is reported with its reason.

Output: pass or reject, before any paid lookup.

04 Contact finder `agents/contact_finder.py, site_emails.py`

Job: Finds one decision-maker and a trustworthy email for them.

- **Decision-makers only:** founders, owners, CEO or president, chiefs, VPs, heads, directors, and managers of marketing-type functions. Associates, coordinators, specialists, analysts and engineers are never returned, and neither are generic inboxes like info@.
- Chooses the target by company size and industry (for example C-suite at tiny startups, marketing leads at larger ones).
- **Step 1 (free):** an email the company itself publishes for a founder, on its own domain, matching the founder's name.
- **Step 2 (optional, paid):** Apify's contact add-on, then a verification check. Off unless `PAID_LOOKUP_FALLBACK=true`.
- **Step 3 (optional):** a guessed address from the founder's name. Off by default; if used it is flagged "guessed" and cannot be sent until confirmed.
- Records every result so discovery learns, and the pipeline cancels any company with no decision-maker email.

Output: name, title, email and LinkedIn link for the contact, plus an email status (published, found, guessed or confirmed).

05 Scoring agent `agents/scoring.py`

Job: Rates how good a customer the company would be, 1 to 10.

- If the tracker or classifier gave a fit rating, that sets the score: Very High 10, High 8, Medium-High 7, Medium 6, Strategic Comparable 3.
- Otherwise it asks Claude (if a key is set) or uses a simple rule score: base 4, plus points for software or e-commerce wording, a small team and funding signals, capped at 10.
- The dashboard counts a lead as "qualified" at 7 or above.

Output: the ICP score and a one or two sentence reason.

06 Research agent `agents/research.py`

Job: Gathers specific, usable observations about the company.

- Reads the company's public website and its YC profile (batch, team size, location, long description, founders).
- Writes two or three observations, lists its sources, and suggests a content angle and best ad type.
- Never uses logins, LinkedIn scraping or anything behind a paywall.

Output: research notes, sources and a recommended outreach angle.

07 Outreach writer `agents/outreach_writer.py`

Job: Drafts the first email in your voice.

- Follows your approved "Allison / Serfis AI" message: what Serfis does and where, why fresh per-platform content matters, an idea for this company, and a closing question about sending a couple of concepts.
- Email only: it never asks for a call, meeting or demo, and never describes itself as an AI pipeline.
- Uses Claude when a key is set (110 to 170 words); otherwise a template that fills in the contact's first name and what the company does.
- Always a **draft**. It cannot send.

Output: an email subject and body saved with the lead, status pending review.

08 Reply writer `agents/reply_writer.py`

Job: Drafts a response when a prospect writes back.

- Only runs when you paste a prospect's reply into the dashboard (there is no live inbox connection yet).
- Answers what the prospect actually said first, in the same voice as the first email.
- If the reply is asking to stop ("unsubscribe", "remove me", "stop emailing"), the address is added to the do-not-contact list and no reply is drafted.

Output: a draft reply waiting for your approval in the lead's Conversation panel.

09 Send agent `agents/send_agent.py`

Job: Sends approved emails through Gmail, with guardrails.

- Never runs as part of discovery. It runs only from the command line: `send` or `send-approved`.
- **Refuses** when the lead is not approved, was already contacted, is on the do-not-contact list, has no usable address (or a guessed, unconfirmed one), is demo data, the daily cap is reached, or the mailing address is not set.
- Every email gets an opt-out line and your postal address in the footer.
- Daily cap (default 20) and a pause between emails (default 45 seconds). Logs each send and marks the lead Contacted.
- Has a dry run that shows who would be emailed without sending, and a test mode that sends one email to an address you choose.

Output: sent emails, a send log, and the lead marked Contacted.

Supporting pieces

Piece	What it does
Orchestrator	Runs the agents in order per company, saves after each step, reports rejected, cancelled and failed companies.
Storage (SQLite)	One database file holds leads, content analytics, conversation messages, the do-not-contact list, the send log and the lookup history.
Excel export	Writes every lead to `output/serfis_leads.xlsx` with all the fields on the lead page.
Website reader	Fetches only public pages, politely, with an identifying user agent.
Apify and Claude wrappers	Shared connections so every agent calls these services the same way.

3. How the dashboard works

The dashboard is where you review leads, approve or reject outreach, and track conversations. It reads the same database the agents write to.

Opening it

From the project folder run `source venv/bin/activate` then `python app.py`, and open **http://127.0.0.1:5050** in a browser. It runs only on your computer.

Home page: the lead list

- **Five cards at the top:** Total Leads, Qualified Leads (score 7 or above), Pending Review, Approved and Rejected.
- **Filters:** search by company name, minimum ICP score, lead status, approval status and industry, and sort by score high to low or low to high.
- **Table:** company, industry, size, ICP score, contact, lead status, approval and date added. Click any row to open that lead.
- **Export to Excel** (top right) downloads the whole list.
- A note appears when the list includes the fictional "Serfis Sample" lead, which exists only to demonstrate the converted-lead view.

Lead page: everything about one company

Panel	What you see
Header	Company, website, ICP score, and badges for lead status and approval status.
Company	Priority and fit, category, market, industry, size, location, what they sell, funding notes, YC batch and profile link, founders.
ICP Fit	Score, why it fits, suggested content angle and best ad type.
Point of Contact	Name, title, email with its status (published, found, guessed or confirmed) and profile link.
Research	Research notes and the sources they came from.
Recommended Outreach Angle	The suggested approach for this company.
Draft Email	The subject and body that will be sent if you approve.
Content Performance Analytics	Views, likes, shares and comments by platform for converted leads. Today only the demo sample lead has data here.
Review This Outreach	**Approve Outreach** and **Reject Outreach** buttons, plus a lead-status menu (New, Contacted, Qualified, Disqualified, Converted).
Conversation	The email thread, any drafted replies with **Approve Reply** and **Reject Reply**, and a form to log a reply from the prospect.

The review workflow

1. Run the pipeline. New leads arrive as **Pending Review**.
2. Open a lead, read the research and the draft email, and check the contact.
3. Click **Approve Outreach** (or Reject). This only records your decision.
4. Send from the command line: `python main.py send-approved`. The lead becomes **Contacted**.
5. When the prospect replies, paste the reply into **Log a reply**. A draft response appears for you to approve or reject.

> **Important:** approving in the dashboard does **not** send anything. Sending is a separate command, so a lead only goes out when you approve it and then run the send step. Approving a reply also does not send it; replies are sent manually for now.

Statuses

Kind	Values	Meaning
Approval	Pending Review, Approved, Rejected	Your decision on the draft outreach. Only Approved leads can be sent.
Lead status	New, Contacted, Qualified, Disqualified, Converted	Where the lead is in the relationship. Contacted is set automatically on send.
Email status	published, found, guessed, confirmed	How the email was obtained. Published = on the company's own site; found = from a data provider; guessed addresses cannot be sent until confirmed.

4. Command reference

Run these from the project folder after `source venv/bin/activate`.

Command	What it does
`python main.py run --limit 5`	Runs the whole pipeline and saves up to 5 new leads for review. Add `--max-lookups N` to cap paid lookups.
`python main.py review`	Prints the leads waiting for approval.
`python main.py approve 12`	Approves lead 12 (the same as the dashboard button).
`python main.py reject 12`	Rejects lead 12.
`python main.py confirm-email 12`	Marks a guessed email as verified; add `--email x@y.com` to correct it.
`python main.py send 12 --to me@x.com`	Sends lead 12's draft to a test address you choose, marked [TEST]. Does not change the lead.
`python main.py send 12`	Sends lead 12's approved draft to the lead.
`python main.py send-approved --dry-run`	Shows who would be emailed. Remove `--dry-run` to send. Add `--limit N` to send fewer.
`python main.py suppress a@b.com`	Adds an address to the do-not-contact list forever.

Command	What it does
`python main.py export`	Writes the Excel file.
`python main.py seed-demo`	Adds the fictional sample lead.
`python main.py reset`	**Deletes all leads** after asking you to confirm. Use with care.

5. Settings (the .env file)

Secrets live in `.env` and are never committed to git.

Setting	Default	Controls
APIFY_API_TOKEN	set	Access to Apify. Without it the system runs in demo mode with seed data.
ANTHROPIC_API_KEY	not set	When set, Claude writes scoring reasons, research and emails. Without it, built-in rules and templates are used.
GMAIL_ADDRESS / GMAIL_APP_PASSWORD	set	The Gmail account emails are sent from (a 16-character app password).
MAILING_ADDRESS	**not set**	The postal address added to every email (legally required). Real sends are blocked until it is set.
EMAIL_DAILY_CAP	20	The most emails sent in a day.
EMAIL_SEND_DELAY_SECONDS	45	Pause between emails.
PAID_LOOKUP_FALLBACK	false	Allow Apify's paid contact lookup when no published founder email is found.
MAX_LOOKUPS_PER_RUN	30	Cap on paid lookups per run.
GUESS_FOUNDER_EMAILS	false	Allow guessed first-name addresses. They stay blocked from sending until confirmed.
APIFY_CONTACT_MAX_CHARGE_USD	0.50	Hard spending cap per paid lookup (Apify's minimum).
DEMO_MODE	false	Forces demo mode with seed companies.
PORT	5050	Port the dashboard runs on.

6. Guardrails and limits

What keeps this safe

- **Human approval first.** Nothing is emailed unless a person approved that lead and then ran the send step.
- **Spend nothing on poor fits.** The ICP guardrail runs before any paid lookup, and a lookup history means no company is paid for twice.
- **Decision-makers only,** with a published or verified email. Guessed addresses are blocked from sending.
- **Respectful sending.** A daily cap, a pause between emails, an opt-out line and postal address in every email, and a permanent do-not-contact list. An unsubscribe reply is honored automatically and never answered with

another pitch.
- **Public data only.** Websites and YC pages are read politely; there is no LinkedIn scraping or login-gated data.

Known limits today

- No live inbox connection: replies are logged by pasting them into the dashboard.
- No scheduler: the pipeline runs when you run it, and the dashboard only while your computer is on.
- Without a Claude API key, emails and research use templates, so they are less tailored.
- Few startups publish a founder email, so a run may return fewer leads than requested; rerun to draw the next ones.
- An email's mailbox cannot be proven to exist without sending, so "published" is the strongest evidence available.

Next steps to make it autonomous

- Set the mailing address and a DMARC record, then start real sends with approval.
- Add automatic bounce and reply detection from the Gmail inbox, and a follow-up agent.
- Run the pipeline daily on a schedule, with a spend cap and failure alerts.
- Auto-send only strictly vetted leads with a daily cap, add a results dashboard, and keep replies human-approved.